LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Steps in aircraft design, manufacturing process and aircraft design requirements
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Steps in aircraft design, manufacturing process and aircraft design requirements
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of aircraft structural elements diagrams; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary structural elements of an aircraft?
(b) List the sequential steps involved in aircraft design.
(c) Define the term 'airworthiness' in the context of design requirements.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Phases of Aircraft Design Process • Aircraft Design Requirements • Structural Elements Identification • Manufacturing Process Overview • Material Selection Criteria	L & Ds	PPT, IB	15 min
(b)	• Detailed Structural Element Analysis • Manufacturing Techniques Discussion • Design Requirement Compliance • Material Properties Review • Integration of Design and Manufacturing	L & Ds	PPT, PDF	20 min
(c)	• Case Study: Design to Manufacturing • Review of Structural Integrity • Summary of Key Requirements • Q&A Session • Preparation for Next Lesson	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between aircraft design requirements and structural elements.
(b) Describe how the manufacturing process impacts the final aircraft structure.

12. Home Assignment and Follow Up	Prepare a brief note on the specific design requirements for the next lesson on Airworthiness and certification requirements of aircraft.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Airworthiness and certification requirements of aircraft	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min